Kitchen Blueprint Pack

Full kitchen planning, equipment schedule & services design — heritage shophouse conversion, Bishop Street, George Town, Penang

PROJECT REF JRT-2026-AGL-01	**PREPARED FOR** Auntie Gaik Lean's Old School Eatery
PREPARED BY JR-Tech Solution Sdn Bhd × TechNext	**DATE** 4 July 2026
REVISION R1	**STATUS** SAMPLE — illustrative deliverable

Document contents

1.0 Scope of this pack

This Blueprint Pack converts one site inspection and one menu workshop into a buildable kitchen design for a **48-seat Nyonya restaurant** trading lunch and dinner (closed Mondays) at an average check of RM95 — a normal day of 55 lunch plus 75 dinner covers, roughly **130 covers ≈ RM12,350/day**. The kitchen occupies **~55 m²** at the rear of a narrow heritage shophouse, with exhaust and deliveries routed to the back lane.

The design is anchored to the menu the kitchen must serve every day: Curry Kapitan, Perut Ikan, Gulai Tumis, Jiu Hu Char, Nyonya Chap Chye, Otak-otak, Kerabu Bee Hoon and Cendol. These dishes define six stations — a two-wok line, simmer/stock, steam, cold prep/kerabu, dessert/plating, and a dishwash return — and every schedule in this document flows from them. The pack covers an equipment investment of **RM185,000**, a **7-week** build timeline, and JR-Tech recurring care at **RM2,980/month** (maintenance + chemicals). Excluded: civil/structural works, front-of-house fit-out, and licence fees payable to authorities.

> This document is an illustrative **sample** prepared by JR-Tech × TechNext to demonstrate the standard blueprint deliverable. All figures are modelled for demonstration and do not constitute a quotation.

2.0 Site & Constraints

The premises is a pre-war shophouse within the George Town UNESCO buffer zone. The building itself is the brand — so the design works entirely within the envelope: nothing penetrates the street facade, all heavy services route to the rear lane, and equipment is sized to pass a **1.1 m** rear door.

2.1 Shophouse constraint register

CONSTRAINT	CONDITION ON SITE	DESIGN RESPONSE
Internal width	Narrow plan, ~4.6 m clear at kitchen bay	Single-corridor galley layout; equipment max 750 mm deep on the wok wall, 700 mm opposite
Rear-lane access	Delivery and waste via back lane; door opening 1.1 m	All equipment specified ≤1,050 mm crated width or knock-down; SS fabrication site-assembled
Heritage facade rules	No new openings, ducts or plant visible from Bishop Street	Exhaust discharges to rear lane only; no roof plant on street slope; signage untouched
Exhaust route	Rear wall to lane, ~9 m duct run from wok line	Smokeless hood with ESP filter so lane discharge is low-smoke, low-odour (Section 6)
Structure	Timber upper floor; no heavy loads on suspended slab	All heavy items (stock pot, dish machine, wok range) on ground-bearing slab at rear
Party walls	Shared masonry both sides	No chasing of party walls; services run in surface SS trunking and floor screed

2.2 Site-inspection summary (services)

ITEM	FINDING	STATUS
Electrical supply	Existing **single-phase 60A**. Connected kitchen load requires **3-phase 60A**; TNB upgrade application to run in parallel with fit-out (Section 5.4)	Upgrade required
Water pressure	1.4 bar at rear tap — adequate for sinks; booster set specified ahead of the hood dishwasher and filtration train	Booster specified

Drainage	Existing fall to rear lane confirmed by dye test; new trapped floor gullies to be laid to the same fall	Adequate
Floor loading	Ground-bearing slab at kitchen bay; localised screed repair at old wet-kitchen area	Adequate
Gas	No piped gas; LPG cylinder bank at rear lane cage with solenoid interlock to hood (Section 6)	LPG design incl.

2.3 Site context

flowchart LR ST["Bishop Street
heritage facade — no works"] --- FOH["Front of house
48 seats, air-well"] FOH --- KIT["Kitchen bay
~55 m², rear of shophouse"] KIT --- RL["Rear lane
1.1 m door"] RL --> DEL["Deliveries
morning window 7–10 am"] RL --> EXH["Exhaust discharge
ESP-filtered, low level"] RL --> LPG["LPG cylinder cage
interlocked"] RL --> WST["Waste + MBPP collection
grease-trap service access"] style ST fill:#f3f3f3,stroke:#777 style KIT fill:#eef4fb,stroke:#0B1F3A style RL fill:#e8f8f3,stroke:#16C098

3.0 Zoned Layout & One-Way Flow

The 55 m² galley enforces a strict one-way food-safety flow — raw goods only ever move forward, from the rear-lane receiving point toward the pass — while soiled ware returns on a separate dishwash loop and waste exits straight to the rear lane without crossing prep or the line. This is the layout logic MBPP food-premises inspectors expect to see.

3.1 Zoned floorplan concept

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flowchart LR RCV["Receiving
rear-lane door 1.1 m"] --> DRY["Dry + cold store
upright freezer, shelving"] DRY --> CP["Cold prep + kerabu
salad-top counters"] CP --> WOK["Wok line
2 woks, smokeless hood"] CP --> STM["Steam + simmer
steamer, stock, induction"] WOK --> PASS["Plating + pass
dessert station"] STM --> PASS PASS --> SVC["Service / dining
48 seats"] SVC -. "soiled ware only" .-> DW["Dishwash return
hood washer + heat recovery"] DW -. "clean ware" .-> PASS DW -.
"clean pots" .-> CP CP -. "covered bins" .-> WR["Waste route
direct to rear lane"] DW -.-> WR WR -.-> RLN["Rear lane
MBPP collection"] style RCV fill:#eef4fb,stroke:#0B1F3A style CP
fill:#eef4fb,stroke:#0B1F3A style WOK fill:#fdeeee,stroke:#a33 style
STM fill:#fdeeee,stroke:#a33 style PASS fill:#e8f8f3,stroke:#16C098
style DW fill:#fff8e6,stroke:#b8860b style WR fill:#f3f3f3,stroke:#777
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3.2 Zone areas (of 55 m²)

ZONE	KEY STATIONS	AREA	SHARE
Receiving + dry/cold store	Rear door, shelving, upright freezer	9.0 m²	16%
Cold prep / kerabu	Salad-top counters, prep sink, rempah bench	8.0 m²	15%
Wok line	Double-wok range, smokeless hood, undercounter chiller	10.0 m²	18%
Steam / simmer / stock	Steamer, RN stock pot, induction simmer	7.0 m²	13%

Plating / pass + dessert	Pass counter, rice/dessert station, cendol chiller	7.0 m²	13%
Dishwash return	Inlet/outlet tables, hood dishwasher	7.0 m²	13%
Waste route + circulation	Bin dock, grease-trap access, corridor	7.0 m²	13%
Total kitchen		**55.0 m²**	**100%**

> The kerabu/cold-prep zone is deliberately placed at the cool front end of the galley, away from the wok line — kerabu bee hoon and cendol assembly are the two most temperature-sensitive items on the menu, and the layout keeps them out of the hood's heat plume.

4.0 Equipment Schedule

Eighteen line items, sized to the menu and the 1.1 m rear door. Point references map to the electrical (P) and drainage (D) schedules on Sheet 5. Indicative pricing totals **RM185,000**; JR-Tech in-house stainless fabrication covers all worktables, sinks, shelving and the dishwash tabling.

ID	ITEM	QTY	ZONE	DIMS (MM)	KW / GAS	ELEC PT	W
E-01	Double-wok range, duplex burners, water-cooled top	1	Wok line	1500×750×800	LPG 2×36 kW	P-01	C
E-02	Smokeless hood with ESP filter + fan, 3.2 m run	1	Wok line	3200×1000×550	3.0	P-02	—
E-03	Induction simmer range, 2-zone (curries, gulai)	1	Steam/simmer	800×750×800	7.0	P-03	—
E-04	3-tier gas steamer cabinet (otak-otak, chap chye)	1	Steam/simmer	900×850×1750	LPG 24 kW	—	C
E-05	RN stock-pot burner, low pedestal (stock, perut ikan)	1	Steam/simmer	600×600×450	LPG 18 kW	—	C
E-06	Undercounter chiller, 2-door GN (wok line mise)	1	Wok line	1360×700×850	0.45	P-04	—

E-07	Undercounter chiller, 2-door GN (cold prep)	1	Cold prep	1360×700×850	0.45	P-05	—
E-08	Upright freezer, 2-door, knock-down delivery	1	Store	1050×800×2000	0.75	P-06	—
E-09	Cold-prep counter with refrigerated salad top (kerabu)	1	Cold prep	1500×700×850	0.50	P-07	—
E-10	Rice/dessert station counter with warmer + gantry	1	Plating/pass	1400×700×850	1.20	P-08	—
E-11	Hood-type dishwasher with heat-recovery condenser	1	Dishwash	650×740×1520	12.60	P-09	C
E-12	SS worktables, 1.2 mm 304, undershelf (4 units)	4	Various	1200/1500×700×850	—	—	—
E-13	SS wall shelving + 4-tier storage racks, site-assembled	1 set	Store / walls	various	—	—	—
E-14	Double-bowl prep sink + 3 hand-wash basins, knee-op	1 set	Cold prep / entries	1500×700×850 + 400 sq	—	CW×4	D 0
E-15	Grease trap, 100 L SS undermount,	1	Waste route	900×500×450	—	—	—

	lane-side access						
E-16	Water filtration + softener train (dishwasher, steamer, drinks)	1	Dishwash	wall-mount	0.20	P-10	C
E-17	Dishwash inlet/outlet tables + pre-rinse spray unit	1 set	Dishwash	1200 + 1000×700	—	—	C
E-18	Dessert display chiller (cendol toppings, gula melaka)	1	Plating/pass	900×570×680	0.35	P-11	—
Total equipment & fabrication investment (± 8%, pre-SST)							

Every powered item above 1 kW carries a dedicated point reference; the remaining spare way **P-12** is reserved at the DB for future cold-room or a-la-minute induction capacity. Recurring care (planned maintenance + warewash chemicals) runs at **RM2,980/month** under the JR-Tech service agreement — detailed in the Handover sample.

SAMPLE

5.0 Electrical & Drainage Point Schedules

5.1 Electrical point schedule (P-01 … P-12)

REF	TYPE	GRID	SERVES	LOAD (KW)	CIRCUIT
P-01	SSO 13A + ignition feed	B2	E-01 wok range controls/ignition	0.30	1P 20A RCBO
P-02	Isolator 20A	B1	E-02 ESP hood + extract fan	3.00	1P 20A RCBO
P-03	Isolator 32A, 3-phase	C2	E-03 induction simmer range	7.00	3P 16A MCB
P-04	SSO 13A	B3	E-06 undercounter chiller (wok)	0.45	1P 13A ring
P-05	SSO 13A	A3	E-07 undercounter chiller (prep)	0.45	1P 13A ring
P-06	SSO 13A	A1	E-08 upright freezer	0.75	1P 13A dedicated
P-07	SSO 13A	A2	E-09 salad-top prep counter	0.50	1P 13A ring
P-08	SSO 13A ×2	D2	E-10 rice/dessert warmer + gantry	1.20	1P 16A RCBO
P-09	Isolator 32A, 3-phase	D3	E-11 hood dishwasher + heat recovery	12.60	3P 25A MCB
P-10	SSO 13A	D3	E-16 filtration/softener + booster pump	0.55	1P 13A ring
P-11	SSO 13A	D1	E-18 dessert display chiller	0.35	1P 13A ring
P-12	Spare way at DB	—	Future cold room / induction	—	3P spare
Connected electrical load (excl. lighting/small power allowance 2.6 kW)				**27.15 + 2.60 = 29.75**	

5.2 Drainage point schedule (D-01 … D-08, GT-01)

REF	TYPE	GRID	SERVES	ROUTE
D-01	Trapped floor gully, SS grate	B2	Wok range water-cooled top	Via GT-01 to rear lane
D-02	Trapped floor gully	C1	Steamer condensate + blowdown	Via GT-01
D-03	Trapped floor gully	C2	Stock-pot station washdown	Via GT-01
D-04	Tundish, 50 mm standpipe	D3	Dishwasher discharge (60–65 °C rated)	Via GT-01
D-05	Waste to trap, 40 mm	A2	Double-bowl prep sink	Via GT-01
D-06	Waste to trap, 32 mm ×3	A2/B1/D3	Hand-wash basins (soap water only)	Direct to foul
D-07	Tundish	D3	Filtration backwash / RO reject	Direct to foul
D-08	Trapped floor gully	D3	Dishwash area washdown	Via GT-01
GT-01	Grease trap 100 L (E-15)	E1 (lane side)	All greasy waste lines	To rear-lane foul; weekly service access from lane

5.3 Connected load by zone

5.4 Load summary & supply upgrade

29.75 kW
CONNECTED LOAD (INCL. LIGHTING ALLOWANCE)

0.65
DIVERSITY FACTOR APPLIED → 19.3 KW DEMAND

3φ 60A
RECOMMENDED SUPPLY (≈ 41 KVA HEADROOM)

Supply upgrade required. The existing **single-phase 60A** supply (~13.8 kW) cannot carry the dishwasher and induction range together. A **TNB upgrade to 3-phase 60A** with a new DB and RCBO board is included in the 7-week programme; the application is lodged in week 1 so the meter change lands before commissioning in week 6. The two heaviest cooking loads (woks, steamer, stock pot) run on LPG, deliberately keeping electrical demand well inside the upgraded supply.

6.0 Ventilation & Statutory Compliance

6.1 Exhaust & hood specification — wok cooking in a heritage shophouse

Two high-output woks in a UNESCO buffer-zone shophouse are the hardest ventilation case in Penang F&B: heavy smoke and oil aerosol, no facade penetrations allowed, and neighbours across a narrow lane. The specification answers all three:

- **Smokeless hood (E-02)** — 3.2 m stainless canopy over the wok line and steam/simmer bank, with integral **electrostatic precipitator (ESP)** removing ≥95% of oil mist and smoke particles, plus an activated-carbon odour stage before discharge.
- **Discharge to rear lane** — low-level louvre on the rear wall, angled away from the neighbouring shophouse; because the air is ESP-cleaned, no unsightly external stack is needed and the heritage facade is untouched.
- **Make-up air** — passive intake via the existing air-well path, keeping the kitchen at slight negative pressure so cooking smells never drift into the 48-seat dining room.
- **Fire safety at the hood** — grease-rated filters, hood fire-suppression drop, and duct access panels at 3 m centres for cleaning contracts under the maintenance agreement.

6.2 Gas (LPG) installation

Cylinder bank in a ventilated lockable cage on the rear lane, manifolded and piped in copper to the wok range, steamer and stock pot. A **solenoid gas interlock** shuts the supply if the exhaust fan stops, with an emergency knock-off button at the kitchen exit — installed and certified by a Suruhanjaya Tenaga-registered gas contractor.

6.3 Compliance checklist

REQUIREMENT	AUTHORITY	ACTION IN THIS PROJECT	STATUS
Food premises licence (new kitchen layout)	MBPP	Layout drawings from Sheet 3 submitted with licence application; inspection at week 7	Submit wk 1
Fire certificate / plans approval	BOMBA	Hood suppression, extinguishers, LPG cage and escape route shown on plan set	Submit wk 1

LPG piping & interlock certification	Suruhanjaya Tenaga (registered contractor)	Installation + pressure test certificate before commissioning	Week 5-6
Electrical supply upgrade to 3-phase 60A	TNB	Application week 1; new DB by licensed electrical contractor	Upgrade required
Typhoid vaccination (TY2) for all food handlers	KKM / clinic	All 9 kitchen staff certified before opening; register kept on site	Before opening
Food-handler training (SLPM)	KKM-accredited trainer	Scheduled during week 6 alongside equipment training	Week 6
Grease trap & waste management	MBPP / IWK	GT-01 sized and positioned for lane-side servicing; service contract in maintenance plan	Designed in
Heritage / facade conformity	George Town World Heritage guidelines	No street-facing works; all plant and discharge at rear lane	Compliant by design
Halal-ready provision (optional, future)	JAKIM	Layout supports segregated storage and colour-coded utensils; no pork/lard on current menu — certification is the owner's future option, not part of this scope	Ready

SAMPLE

7.0 Revisions & Acceptance

7.1 Revision workflow

This pack is issued as **Revision R1**. Markups from the owner or her architect are returned to JR-Tech through the shared project workspace; because the menu, equipment schedule, layout and services points live in one model, any change re-flows automatically through every schedule and the budget. A revised pack is re-issued within **48 hours** of receiving consolidated comments. The pack proceeds to fabrication and M&E first-fix only after signature below.

REV	DATE	DESCRIPTION	ISSUED BY
R1	4 July 2026	First issue for owner review — full blueprint pack	JR-Tech design office
R2	—	Reserved: owner/architect markups incorporated (48 h re-issue)	—
R3	—	Reserved: construction issue (signed)	—

7.2 Acceptance

Signature below confirms acceptance of the zoned layout (Sheet 3), the equipment schedule and RM185,000 indicative budget (Sheet 4), the services design and 3-phase 60A upgrade (Sheet 5), and the compliance pathway (Sheet 6), and authorises JR-Tech to proceed with the 7-week programme.

For the Owner
Auntie Gaik Lean's Old School Eatery

NAME & SIGNATURE

DATE

For JR-Tech Solution Sdn Bhd
Project Director, JRT-2026-AGL-01

NAME & SIGNATURE

DATE

> Companion samples in this series: Operations Pack · Financial Pack · Handover Pack — all for the same illustrative engagement.